# Technical Note: Design Theory for the Butterworth-Optimized Photodiode Transimpedance Amplifier

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**Abstract**

This document provides a self-contained description of the mathematical theory implemented in the photodiode transimpedance amplifier (TIA) design tool. It covers the single-pole opamp model, the derivation of the closed-loop second-order transfer function, the exact Butterworth-optimum feedback capacitance without the common $\omega_c R_f C_f \gg 1$ approximation, the $-3\,\mathrm{dB}$ bandwidth, the noise equivalent bandwidth for a second-order Butterworth response, the piecewise noise-gain integration for opamp voltage noise, and the shot/thermal noise analysis. All algebraic steps are shown explicitly.

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# 1 Circuit model and assumptions

## 1.1 Circuit topology

The TIA is an inverting opamp configuration with a photodiode at the input. The non-inverting input is connected to ground.

- $I_{\mathrm{pd}}$: photocurrent from the photodiode (signal source, modelled as ideal current source)
- $R_{\mathrm{f}}$, $C_{\mathrm{f}}$: feedback resistance and capacitance (in parallel), connected from the output to the inverting input
- $C_{\mathrm{i}}$: total input capacitance at the inverting node, $C_{\mathrm{i}} = C_{\mathrm{d}} + C_{\mathrm{icm}} + C_{\mathrm{id}}$
  - $C_{\mathrm{d}}$: photodiode junction capacitance
  - $C_{\mathrm{icm}}$: opamp common-mode input capacitance
  - $C_{\mathrm{id}}$: opamp differential input capacitance
- $C_{\mathrm{s}}$: stray (parasitic) capacitance at the inverting node to ground

## 1.2 Opamp model

The opamp is modelled with a single-pole open-loop gain:

$$A(s) = \frac{A_0\,\omega_{\mathrm{p}}}{s + \omega_{\mathrm{p}}} \tag{1}$$

where $A_0$ is the DC open-loop gain and $\omega_{\mathrm{p}}$ is the dominant pole frequency. For frequencies well above the dominant pole ($s \gg \omega_{\mathrm{p}}$, i.e. $f \gg f_{\mathrm{p}}$, typically satisfied for $f > 10\,\mathrm{Hz}$), this simplifies to:

$$A(s) \approx \frac{A_0\,\omega_{\mathrm{p}}}{s} = \frac{\omega_{\mathrm{c}}}{s} \tag{2}$$

where $\omega_{\mathrm{c}} = A_0\,\omega_{\mathrm{p}} = 2\pi f_{\mathrm{c}}$ is the gain-bandwidth product (GBW) in $\mathrm{rad\,s^{-1}}$. This "integrator model" is the standard approximation used throughout TIA design literature and is accurate across the entire frequency range of interest (kHz–MHz).

# 2 Closed-loop transimpedance transfer function

## 2.1 Goal

Derive the closed-loop transimpedance:

$$Z_{\mathrm{TI}}(s) \equiv \frac{V_{\mathrm{out}}(s)}{I_{\mathrm{pd}}(s)}.$$

## 2.2 Step 1: Define the feedback impedance

$R_{\mathrm{f}}$ and $C_{\mathrm{f}}$ are in parallel:

$$Z_{\mathrm{f}} = R_{\mathrm{f}} \parallel \frac{1}{sC_{\mathrm{f}}} = \frac{R_{\mathrm{f}}}{1 + sR_{\mathrm{f}}C_{\mathrm{f}}} \tag{3}$$

## 2.3 Step 2: Opamp constraint

With the non-inverting input grounded:

$$V_{\mathrm{out}} = -A(s) \cdot V^- = -\frac{\omega_{\mathrm{c}}}{s}\,V^-$$

Solving for $V^-$:

$$V^- = -\frac{s}{\omega_{\mathrm{c}}}\,V_{\mathrm{out}} \tag{4}$$

### 2.4 Step 3: Kirchhoff's current law at the inverting node

All currents leaving the inverting node must sum to zero. The current into the node from the photodiode is $I_{\mathrm{pd}}$. The currents leaving the node are:

- Through the feedback network to the output: $(V^- - V_{\mathrm{out}})/Z_{\mathrm{f}}$
- Through the total shunt capacitance $(C_{\mathrm{i}} + C_{\mathrm{s}})$ to ground: $V^- \cdot s(C_{\mathrm{i}} + C_{\mathrm{s}})$

KCL:

$$I_{\mathrm{pd}} + \frac{V^- - V_{\mathrm{out}}}{Z_{\mathrm{f}}} + V^- \cdot s(C_{\mathrm{i}} + C_{\mathrm{s}}) = 0$$

Note the sign convention: current flowing from the node through $Z_{\mathrm{f}}$ toward $V_{\mathrm{out}}$ is $(V^- - V_{\mathrm{out}})/Z_{\mathrm{f}}$. Rearranging with current into the node on the left:

$$I_{\mathrm{pd}} = \frac{V_{\mathrm{out}} - V^-}{Z_{\mathrm{f}}} - V^- \cdot s(C_{\mathrm{i}} + C_{\mathrm{s}}) \tag{5}$$

### 2.5 Step 4: Substitute the opamp constraint

Substitute (3) and (4) into (5).

First, compute $V_{\mathrm{out}} - V^-$:

$$V_{\mathrm{out}} - V^- = V_{\mathrm{out}}\left(1 + \frac{s}{\omega_{\mathrm{c}}}\right) = V_{\mathrm{out}} \cdot \frac{\omega_{\mathrm{c}} + s}{\omega_{\mathrm{c}}} \tag{6}$$

Substitute into the feedback current term:

$$\frac{V_{\mathrm{out}} - V^-}{Z_{\mathrm{f}}} = V_{\mathrm{out}} \cdot \frac{\omega_{\mathrm{c}} + s}{\omega_{\mathrm{c}}} \cdot \frac{1 + sR_{\mathrm{f}}C_{\mathrm{f}}}{R_{\mathrm{f}}} \tag{7}$$

The shunt capacitance current term:

$$-V^- \cdot s(C_{\mathrm{i}} + C_{\mathrm{s}}) = \frac{s^2(C_{\mathrm{i}} + C_{\mathrm{s}})}{\omega_{\mathrm{c}}} V_{\mathrm{out}} \tag{8}$$

### 2.6 Step 5: Combine

Substituting (7) and (8) into (5):

$$I_{\mathrm{pd}} = V_{\mathrm{out}} \left[\frac{(\omega_{\mathrm{c}} + s)(1 + sR_{\mathrm{f}}C_{\mathrm{f}})}{\omega_{\mathrm{c}}R_{\mathrm{f}}} + \frac{s^2(C_{\mathrm{i}} + C_{\mathrm{s}})}{\omega_{\mathrm{c}}}\right]$$

Factor out $1/\omega_{\mathrm{c}}$:

$$I_{\mathrm{pd}} = \frac{V_{\mathrm{out}}}{\omega_{\mathrm{c}}} \left[\frac{(\omega_{\mathrm{c}} + s)(1 + sR_{\mathrm{f}}C_{\mathrm{f}})}{R_{\mathrm{f}}} + s^2(C_{\mathrm{i}} + C_{\mathrm{s}})\right] \tag{9}$$

### 2.7 Step 6: Expand the product

$$\begin{aligned}(\omega_{\mathrm{c}} + s)(1 + sR_{\mathrm{f}}C_{\mathrm{f}}) &= \omega_{\mathrm{c}} + \omega_{\mathrm{c}}sR_{\mathrm{f}}C_{\mathrm{f}} + s + s^2R_{\mathrm{f}}C_{\mathrm{f}} \\ &= s^2R_{\mathrm{f}}C_{\mathrm{f}} + s(1 + \omega_{\mathrm{c}}R_{\mathrm{f}}C_{\mathrm{f}}) + \omega_{\mathrm{c}}\end{aligned}$$

### 2.8 Step 7: Divide by $R_{\mathrm{f}}$ and add the shunt term

$$\begin{aligned}&\frac{(\omega_{\mathrm{c}} + s)(1 + sR_{\mathrm{f}}C_{\mathrm{f}})}{R_{\mathrm{f}}} + s^2(C_{\mathrm{i}} + C_{\mathrm{s}}) \\ &= s^2C_{\mathrm{f}} + \frac{s(1 + \omega_{\mathrm{c}}R_{\mathrm{f}}C_{\mathrm{f}})}{R_{\mathrm{f}}} + \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}} + s^2(C_{\mathrm{i}} + C_{\mathrm{s}}) \\ &= s^2(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}) + s\,\frac{1 + \omega_{\mathrm{c}}R_{\mathrm{f}}C_{\mathrm{f}}}{R_{\mathrm{f}}} + \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}}\end{aligned}$$

### 2.9 Step 8: Form the transfer function

From (9) and the result of Step 7:

$$Z_{\mathrm{TI}}(s) = \frac{V_{\mathrm{out}}}{I_{\mathrm{pd}}} = \frac{\omega_{\mathrm{c}}}{s^2(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}) + s\,\dfrac{1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}} + \dfrac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}}} \tag{10}$$

(The overall sign is negative for the inverting configuration; we write the magnitude here and note the 180° phase inversion separately.)

### 2.10 Step 9: Identify the standard second-order form

A standard second-order low-pass transfer function has the form:

$$H(s) = \frac{K\,\omega_{\mathrm{n}}^2}{s^2 + 2\zeta\omega_{\mathrm{n}}\,s + \omega_{\mathrm{n}}^2}$$

Dividing numerator and denominator of (10) by $(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})$:

$$Z_{\mathrm{TI}}(s) = \frac{\dfrac{\omega_{\mathrm{c}}}{C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}}}{s^2 + \dfrac{1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}\,s + \dfrac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}} \tag{11}$$

By comparison with the standard form, we identify:

$$\omega_{\mathrm{n}}^2 = \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})} \tag{12}$$

$$2\zeta\omega_{\mathrm{n}} = \frac{1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})} \tag{13}$$

The DC gain ($s = 0$) of (11):

$$Z_{\mathrm{TI}}(0) = \frac{\omega_{\mathrm{c}}/(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}{\omega_{\mathrm{c}}/[R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})]} = R_{\mathrm{f}}.$$

The numerator constant can therefore be written as $R_{\mathrm{f}}\,\omega_{\mathrm{n}}^2$, giving the final form (restoring the sign for the inverting configuration):

$$Z_{\mathrm{TI}}(s) = \frac{-R_{\mathrm{f}}\,\omega_{\mathrm{n}}^2}{s^2 + 2\zeta\omega_{\mathrm{n}}\,s + \omega_{\mathrm{n}}^2} \tag{14}$$

## 3 Butterworth-optimum feedback capacitance

### 3.1 Objective

Find the value of $C_{\mathrm{f}}$ that makes the closed-loop response maximally flat (Butterworth).

### 3.2 Step 1: Butterworth condition

A second-order Butterworth (maximally flat magnitude) filter has quality factor:

$$Q = \frac{1}{\sqrt{2}}$$

The relationship between $Q$ and the damping ratio $\zeta$ is $Q = 1/(2\zeta)$, so:

$$\zeta = \frac{1}{\sqrt{2}} \tag{15}$$

This is the unique damping ratio that produces zero gain peaking in the magnitude response (i.e. $|Z_{\mathrm{TI}}(j\omega)|$ is monotonically decreasing).

### 3.3 Step 2: Translate the condition to circuit parameters

The Butterworth condition (15) implies:

$$(2\zeta\omega_{\mathrm{n}})^2 = 4\zeta^2\omega_{\mathrm{n}}^2 = 4 \cdot \frac{1}{2} \cdot \omega_{\mathrm{n}}^2 = 2\,\omega_{\mathrm{n}}^2.$$

Substituting (12) and (13):

$$\left(\frac{1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}\right)^2 = 2 \cdot \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}$$

Multiply both sides by $[R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})]^2$:

$$(1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}})^2 = 2\,\omega_{\mathrm{c}} R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}) \tag{16}$$

### 3.4 Step 3: Solve for $C_{\mathrm{f}}$

Define a characteristic capacitance:

$$C_{\mathrm{c}} \equiv \frac{1}{\omega_{\mathrm{c}} R_{\mathrm{f}}} = \frac{1}{2\pi f_{\mathrm{c}} R_{\mathrm{f}}} \tag{17}$$

This has a physical interpretation: $C_{\mathrm{c}}$ is the capacitance whose impedance at $f_{\mathrm{c}}$ equals $R_{\mathrm{f}}$.

Rewrite (16) using $\omega_{\mathrm{c}} R_{\mathrm{f}} = 1/C_{\mathrm{c}}$:

$$\left(1 + \frac{C_{\mathrm{f}}}{C_{\mathrm{c}}}\right)^2 = \frac{2(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}{C_{\mathrm{c}}}$$

Expand the left side:

$$1 + \frac{2C_{\mathrm{f}}}{C_{\mathrm{c}}} + \frac{C_{\mathrm{f}}^2}{C_{\mathrm{c}}^2} = \frac{2(C_{\mathrm{i}} + C_{\mathrm{s}})}{C_{\mathrm{c}}} + \frac{2C_{\mathrm{f}}}{C_{\mathrm{c}}}$$

The $2C_{\mathrm{f}}/C_{\mathrm{c}}$ terms cancel:

$$1 + \frac{C_{\mathrm{f}}^2}{C_{\mathrm{c}}^2} = \frac{2(C_{\mathrm{i}} + C_{\mathrm{s}})}{C_{\mathrm{c}}}$$

Multiply through by $C_{\mathrm{c}}^2$:

$$C_{\mathrm{f}}^2 = 2(C_{\mathrm{i}} + C_{\mathrm{s}})\,C_{\mathrm{c}} - C_{\mathrm{c}}^2 = C_{\mathrm{c}}\big(2(C_{\mathrm{i}} + C_{\mathrm{s}}) - C_{\mathrm{c}}\big)$$

Taking the positive root:

$$C_{\mathrm{f}} = \sqrt{C_{\mathrm{c}}\big(2(C_{\mathrm{i}} + C_{\mathrm{s}}) - C_{\mathrm{c}}\big)}, \qquad C_{\mathrm{c}} = \frac{1}{2\pi f_{\mathrm{c}} R_{\mathrm{f}}} \tag{18}$$

### 3.5 Validity condition

(18) requires the argument of the square root to be non-negative:

$$2(C_{\mathrm{i}} + C_{\mathrm{s}}) - C_{\mathrm{c}} > 0 \quad \Longleftrightarrow \quad C_{\mathrm{i}} + C_{\mathrm{s}} > \frac{C_{\mathrm{c}}}{2} = \frac{1}{2\omega_{\mathrm{c}} R_{\mathrm{f}}}.$$

When this condition is violated, the opamp GBW is too low (or $R_{\mathrm{f}}$ too small) to achieve a Butterworth response—the circuit is fundamentally underdamped regardless of $C_{\mathrm{f}}$.

### 3.6 Relation to the common approximation

Many references simplify by assuming $\omega_c R_f C_f \gg 1$ (equivalently $C_f \gg C_c$). Under this approximation, (16) becomes:

$$(\omega_c R_f C_f)^2 \approx 2\,\omega_c R_f (C_i + C_s) \quad \Longrightarrow \quad C_f \approx \sqrt{\frac{2(C_i + C_s)}{\omega_c R_f}} = \sqrt{2(C_i + C_s)\,C_c}.$$

(18) is the **exact** solution that does not rely on this approximation. The difference is significant when $C_c$ is not negligible compared to $C_i + C_s$ (i.e. when $R_f$ is small or $f_c$ is low).

## 4 Bandwidth

### 4.1 The $-3\,\mathrm{dB}$ bandwidth of a second-order Butterworth

For a general second-order transfer function with damping ratio $\zeta$, the squared magnitude response is:

$$|H(j\omega)|^2 = \frac{\omega_n^4}{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}$$

For the Butterworth case $\zeta = 1/\sqrt{2}$, substituting and simplifying:

$$|H(j\omega)|^2 = \frac{\omega_n^4}{(\omega_n^2 - \omega^2)^2 + 2\omega_n^2\omega^2}$$

Expand the denominator:

$$= \omega_n^4 - 2\omega_n^2\omega^2 + \omega^4 + 2\omega_n^2\omega^2 = \omega_n^4 + \omega^4$$

Therefore:

$$|H(j\omega)|^2 = \frac{\omega_n^4}{\omega_n^4 + \omega^4} = \frac{1}{1 + (\omega/\omega_n)^4} \tag{19}$$

This is the defining form of a second-order Butterworth response.

At $\omega = \omega_n$:

$$|H(j\omega_n)|^2 = \frac{1}{2} \quad \Longrightarrow \quad |H(j\omega_n)| = \frac{1}{\sqrt{2}}\ (-3\,\mathrm{dB}).$$

Therefore the $-3\,\mathrm{dB}$ bandwidth is exactly the natural frequency:

$$\omega_{-3\,\mathrm{dB}} = \omega_n.$$

In terms of ordinary frequency:

$$f_{-3\,\mathrm{dB}} = \frac{\omega_n}{2\pi} \tag{20}$$

### 4.2 Express in terms of circuit parameters

From (12):

$$\omega_n = \sqrt{\frac{\omega_c}{R_f(C_i + C_f + C_s)}} = \sqrt{\frac{2\pi f_c}{R_f(C_i + C_f + C_s)}}$$

Therefore:

$$f_{-3\,\mathrm{dB}} = \frac{\omega_n}{2\pi} = \sqrt{\frac{f_c}{2\pi R_f(C_i + C_f + C_s)}} \tag{21}$$

# 5 Noise equivalent bandwidth

## 5.1 Definition

The noise equivalent bandwidth (NEB) of a filter $H(f)$ is the bandwidth of an ideal rectangular (brick-wall) filter with the same peak gain that passes the same total noise power:

$$\mathrm{NEB} = \frac{1}{|H_{\max}|^2} \int_0^\infty |H(f)|^2 \, \mathrm{d}f$$

For a Butterworth filter, $|H_{\max}| = |H(0)| = 1$, so:

$$\mathrm{NEB} = \int_0^\infty |H(f)|^2 \, \mathrm{d}f \tag{22}$$

## 5.2 General $n$-th order Butterworth NEB

For an $n$-th order Butterworth filter, the squared magnitude response is:

$$|H(f)|^2 = \frac{1}{1 + (f/f_{-3\,\mathrm{dB}})^{2n}}$$

Substituting into (22) with $u = f/f_{-3\,\mathrm{dB}}$:

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \int_0^\infty \frac{\mathrm{d}u}{1 + u^{2n}} \tag{23}$$

The integral $I_n = \displaystyle\int_0^\infty \frac{\mathrm{d}u}{1 + u^{2n}}$ is a standard result that can be evaluated using the substitution $t = u^{2n}$ and recognizing the beta function.

Setting $t = u^{2n}$, so $u = t^{1/(2n)}$, $\mathrm{d}u = \frac{1}{2n} t^{1/(2n)-1} \, \mathrm{d}t$:

$$I_n = \int_0^\infty \frac{1}{1+t} \cdot \frac{1}{2n} t^{1/(2n)-1} \, \mathrm{d}t = \frac{1}{2n} B\left(\frac{1}{2n}, \, 1 - \frac{1}{2n}\right)$$

where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ is the beta function.

Using the reflection formula $\Gamma(z)\,\Gamma(1-z) = \pi/\sin(\pi z)$ with $z = 1/(2n)$:

$$I_n = \frac{1}{2n} \cdot \frac{\pi}{\sin\left(\frac{\pi}{2n}\right)} \tag{24}$$

Therefore:

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2n \, \sin\left(\frac{\pi}{2n}\right)} \tag{25}$$

## 5.3 Specialization to $n = 2$

For the second-order Butterworth ($n = 2$):

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{4 \cdot \frac{\sqrt{2}}{2}} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2\sqrt{2}} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi\sqrt{2}}{4}$$

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi\sqrt{2}}{4} \approx 1.111 \, f_{-3\,\mathrm{dB}} \tag{26}$$

### 5.4 Comparison with the single-pole approximation

For a single-pole (first-order) system ($n = 1$):

$$\mathrm{NEB}_{n=1} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2\sin(\pi/2)} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2} \approx 1.571\, f_{-3\,\mathrm{dB}}.$$

Many TIA references use this single-pole approximation even when the circuit has a second-order response. The second-order Butterworth NEB is $1.111/1.571 = 0.707$ times the single-pole value—a 29% difference that directly affects computed noise levels.

## 6 Noise analysis

The TIA output noise is the root-sum-square (RSS) of several independent noise sources. Each source has a spectral density that is shaped by the closed-loop gain before appearing at the output. The noise sources fall into two categories: (a) voltage noise of the opamp, which is amplified by the frequency-dependent noise gain, and (b) current/thermal noise sources whose spectra are flat and are shaped only by the closed-loop transimpedance $|Z_{\mathrm{TI}}(f)|^2$.

### 6.1 Noise gain

The noise gain $G_{\mathrm{n}}(f)$ is the closed-loop gain from the opamp's input voltage noise source $e_{\mathrm{ni}}$ to the output. For a TIA with feedback impedance $Z_{\mathrm{f}} = R_{\mathrm{f}} \| (1/sC_{\mathrm{f}})$ and total input-to-ground capacitance $(C_{\mathrm{i}} + C_{\mathrm{s}})$, the noise gain is:

$$G_{\mathrm{n}}(f) = 1 + \frac{Z_{\mathrm{in}}}{Z_{\mathrm{f}}} = 1 + \frac{C_{\mathrm{i}} + C_{\mathrm{s}}}{C_{\mathrm{f}} + C_{\mathrm{s}}} \cdot \frac{1 + sR_{\mathrm{f}}(C_{\mathrm{f}} + C_{\mathrm{s}})}{1 + sR_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}$$

This gain exhibits four distinct frequency regions that are exploited for piecewise integration:

1. **Low-frequency flat region** ($f < f_{\mathrm{zf}}$): $G_{\mathrm{n}} \approx 1$
2. **Rising region** ($f_{\mathrm{zf}} < f < f_{\mathrm{pf}}$): $G_{\mathrm{n}} \approx f/f_{\mathrm{zf}}$
3. **Plateau region** ($f_{\mathrm{pf}} < f < f_{\mathrm{i}}$): $G_{\mathrm{n}} \approx 1 + C_{\mathrm{i}}/(C_{\mathrm{f}} + C_{\mathrm{s}})$
4. **Roll-off region** ($f > f_{\mathrm{i}}$): $G_{\mathrm{n}}$ decreases as the loop gain falls below unity

where the characteristic frequencies are:

$$f_{\mathrm{zf}} = \frac{1}{2\pi R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})} \tag{27}$$

$$f_{\mathrm{pf}} = \frac{1}{2\pi R_{\mathrm{f}}(C_{\mathrm{f}} + C_{\mathrm{s}})} \tag{28}$$

$$f_{\mathrm{i}} = f_{\mathrm{c}} \cdot \frac{C_{\mathrm{f}} + C_{\mathrm{s}}}{C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}} \tag{29}$$

Here $f_{\mathrm{zf}}$ is the noise-gain zero frequency, $f_{\mathrm{pf}}$ is the noise-gain pole frequency, and $f_{\mathrm{i}}$ is the unity loop-gain (intersection) frequency.

### 6.2 Opamp voltage noise (piecewise integration)

The opamp input voltage noise spectral density has a $1/f$ component below the corner frequency $f_\mathrm{f}$ and a white (flat) floor $e_\mathrm{nif}$ above it:

$$e_\mathrm{ni}(f) = e_\mathrm{nif}\sqrt{1 + f_\mathrm{f}/f} \tag{30}$$

The output-referred voltage noise RMS is obtained by integrating the product of $e_\mathrm{ni}^2(f)$ and $G_\mathrm{n}^2(f)$ over frequency, divided into five segments. Defining a low-frequency cutoff $f_1$ (set to 0.01 Hz in the implementation):

**Region 1**—$1/f$ noise ($f_1$ to $f_\mathrm{f}$), noise gain $\approx 1$:

$$E_\mathrm{noe,1}^2 = e_\mathrm{nif}^2 \cdot f_\mathrm{f} \ln\left(\frac{f_\mathrm{f}}{f_1}\right) \tag{31}$$

**Region 2**—white noise, flat gain ($f_\mathrm{f}$ to $f_\mathrm{zf}$), noise gain $\approx 1$:

$$E_\mathrm{noe,2}^2 = e_\mathrm{nif}^2 (f_\mathrm{zf} - f_\mathrm{f}) \tag{32}$$

**Region 3**—white noise, rising gain ($f_\mathrm{zf}$ to $f_\mathrm{pf}$), noise gain $\approx f/f_\mathrm{zf}$:

$$E_\mathrm{noe,3}^2 = \frac{e_\mathrm{nif}^2}{f_\mathrm{zf}^2} \cdot \frac{f_\mathrm{pf}^3 - f_\mathrm{zf}^3}{3} \tag{33}$$

**Region 4**—white noise, plateau ($f_\mathrm{pf}$ to $f_\mathrm{i}$), noise gain $\approx 1 + C_\mathrm{i}/(C_\mathrm{f} + C_\mathrm{s})$:

$$E_\mathrm{noe,4}^2 = e_\mathrm{nif}^2 \left(1 + \frac{C_\mathrm{i}}{C_\mathrm{f} + C_\mathrm{s}}\right)^2 (f_\mathrm{i} - f_\mathrm{pf}) \tag{34}$$

**Region 5**—above unity loop gain ($f > f_\mathrm{i}$), gain $\approx f_\mathrm{c}/f$:

$$E_\mathrm{noe,5}^2 = e_\mathrm{nif}^2 \cdot \frac{f_\mathrm{c}^2}{f_\mathrm{i}} \tag{35}$$

Note: (35) arises from $\int_{f_\mathrm{i}}^{\infty} (f_\mathrm{c}/f)^2 \,\mathrm{d}f = f_\mathrm{c}^2/f_\mathrm{i}$.

The total output-referred opamp voltage noise is:

$$E_\mathrm{noe} = \sqrt{E_\mathrm{noe,1}^2 + E_\mathrm{noe,2}^2 + E_\mathrm{noe,3}^2 + E_\mathrm{noe,4}^2 + E_\mathrm{noe,5}^2} \tag{36}$$

### 6.3 Feedback resistor thermal noise

The Johnson noise of $R_\mathrm{f}$ produces a current noise density $\sqrt{4k_\mathrm{B}T/R_\mathrm{f}}$, which flows through $R_\mathrm{f}$ to produce output voltage noise. Since this noise source sees the same transimpedance transfer function as the signal, the NEB ((26)) is the appropriate bandwidth:

$$E_\mathrm{noR} = \sqrt{4k_\mathrm{B}TR_\mathrm{f} \cdot \mathrm{NEB}} \tag{37}$$

### 6.4 Shot noise

Shot noise from DC currents (bias current $I_\mathrm{b}$, dark current $I_\mathrm{d}$, and signal photocurrent $I_\mathrm{p}$) produces white current noise with spectral density $\sqrt{2qI}$. Each is converted to output voltage noise by the transimpedance $R_\mathrm{f}$ over the NEB:

$$E_\mathrm{noi,bias} = R_\mathrm{f}\sqrt{2q\, I_\mathrm{b} \cdot \mathrm{NEB}} \tag{38}$$

$$E_\mathrm{noi,dark} = R_\mathrm{f}\sqrt{2q\, I_\mathrm{d} \cdot \mathrm{NEB}} \tag{39}$$

$$E_\mathrm{noi,sig} = R_\mathrm{f}\sqrt{2q\, I_\mathrm{p} \cdot \mathrm{NEB}} \tag{40}$$

### 6.5 Opamp current noise

The opamp input current noise floor $i_{\mathrm{nif}}$ (white spectral density) also flows through $R_{\mathrm{f}}$:

$$E_{\mathrm{noi,nif}} = R_{\mathrm{f}}\, i_{\mathrm{nif}} \sqrt{\mathrm{NEB}} \tag{41}$$

### 6.6 Total noise and signal-to-noise ratio

The total background noise (all sources except signal shot noise) is:

$$E_{\mathrm{no,bg}} = \sqrt{E_{\mathrm{noe}}^2 + E_{\mathrm{noR}}^2 + E_{\mathrm{noi,bias}}^2 + E_{\mathrm{noi,dark}}^2 + E_{\mathrm{noi,nif}}^2} \tag{42}$$

The signal voltage is $V_{\mathrm{sig}} = R_{\mathrm{f}} \cdot I_{\mathrm{p}} = R_{\mathrm{f}} \cdot r_{\phi} \cdot P_{\mathrm{opt}}$, where $r_{\phi}$ is the photodiode responsivity and $P_{\mathrm{opt}}$ is the incident optical power. The signal-to-noise ratio is:

$$\mathrm{SNR} = \frac{V_{\mathrm{sig}}}{\sqrt{E_{\mathrm{no,bg}}^2 + E_{\mathrm{noi,sig}}^2}} \tag{43}$$

The ratio $E_{\mathrm{no,bg}}/E_{\mathrm{noi,sig}}$ indicates whether the system is shot-noise limited (ratio $< 1$) or background-noise limited (ratio $> 1$).

## 7 Implementation mapping

| Theory | Implementation | File |
|---|---|---|
| Opamp model, (2) | `Opamp` dataclass (`f_c` field) | `tia_design.py` |
| Input capacitance $C_{\mathrm{i}}$ | `C_i = C_d + opamp.C_icm + opamp.C_id` | `tia_design.py` |
| Butterworth $C_{\mathrm{f}}$, (18) | `C_f = (C_c * (2*(C_i+C_s) - C_c))**0.5` | `tia_design.py` |
| Bandwidth, (21) | `BW_t = (f_c / (2*pi*R_f*C_tot))**0.5` | `tia_design.py` |
| NEB, (26) | `NEB = (pi*sqrt(2)/4) * BW_t` | `tia_design.py` |
| Noise gain frequencies, (27)–(29) | `f_zf`, `f_pf`, `f_i` | `tia_design.py` |
| Piecewise voltage noise, (31)–(36) | `E_noe1` ... `E_noe5`, `E_noe` | `tia_design.py` |
| Thermal noise, (37) | `E_noR` | `tia_design.py` |
| Shot noise, (38)–(40) | `E_noi_bias`, `E_noi_dark`, `E_noi_sig` | `tia_design.py` |
| Current noise, (41) | `E_noi_nif` | `tia_design.py` |
| SNR, (43) | `snr` | `tia_design.py` |
| Photodiode $C_{\mathrm{d}}(V_{\mathrm{b}})$ model | `Photodiode.C_d(Vb)` | `tia_design.py` |
| Multi-opamp comparison | `compare_opamps(...)` | `tia_design.py` |

# 8 References


1. J. Graeme, *Photodiode Amplifiers: Op Amp Solutions.* McGraw-Hill, 1996.
2. P. C. D. Hobbs, *Building Electro-Optical Systems: Making It All Work*, 2nd ed. Wiley, 2009.



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